

RESEARCH ARTICLE

Water restriction causes an intergenerational trade-off and delayed mother–offspring conflict in a viviparous lizard

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Abstract

1. Parenting is costly and because the relationship between the mother and embryos is not mutualistic, mother–offspring conflicts may exist whenever resource are scarce. However, intergenerational trade-offs and conflicts resulting from limited access to water, a vital and depreciable resource, remain largely overlooked.
2. In this study, we examined the physiological, reproductive and life-history responses to water restriction in the European Common Lizard (*Zootoca vivipara*). We hypothesized that, under water-limited conditions, pregnant females experience both short-term and long-term physiological impacts (dehydration and stress) underlying an allocation trade-off for water between mothers and offspring.
3. Water restriction led to a decrease in body mass, and an increase in plasma osmolality (dehydration) and corticosterone concentration in both males and females. The extent of the dehydration was positively correlated with fecundity in females. This suggests a trade-off between maternal water balance and allocation of water to developing embryos during reproduction.
4. Water restriction had no immediate effect on reproductive output or offspring morphology at birth. Yet, water restriction in pregnant females enhanced their reproductive effort the following year but reduced the early life growth and annual survival of their second-year offspring.
5. These delayed fitness responses to water restriction in offspring and mothers suggest that water can trigger intergenerational conflicts as demonstrated for energy. Although the mediation of this conflict remains to be clarified, we hypothesized that it represents a selective force that influences reproductive strategies.

KEYWORDS

conflict, corticosterone, dehydration, growth rate, reproductive effort, survival, trade-off

1 | INTRODUCTION

Life-history theory predicts that parents should optimize resource allocation to reproduction (i.e. reproductive effort) in a dynamic manner according to individual state and environmental conditions (McNamara & Houston, 1996; Senner, Conklin, & Piersma, 2015). This is because a disproportionally high current reproductive effort

may compromise future survival and/or reproductive success in iteroparous species due to energetic and other physiological constraints (Harshman & Zera, 2007). The costs and benefits of a high reproductive effort may depend on individual condition, age, resource availability, or previous life-history decisions of parents leading to dynamic, state-dependent optimization of the reproductive effort (McNamara & Houston, 1996; Senner et al., 2015). In addition, because parents

are not perfectly genetically related to their offspring in sexual species (Trivers, 1974), the evolutionary interests of parents and offspring towards parental investment may differ, leading to antagonistic selection of traits related to parental allocation versus offspring acquisition (Crespi & Semeniuk, 2004; Hinde, Johnstone, & Kilner, 2010; Pollux, Meredith, Springer, Garland, & Reznick, 2014; Schrader & Travis, 2008). Intergenerational trade-offs between mother allocation to reproduction and offspring acquisition can occur during embryonic development or egg formation when resources are limited. Ultimately, such trade-offs can result in parent-offspring conflicts over resources as soon as genetic trade-offs between offspring performance and the residual fitness of parents exist (Godfray, Partridge, & Harvey, 1991; Hinde et al., 2010; Kölliker et al., 2015; Parker & Mock, 1987).

Although life-history theory is not explicit about the resource involved in the battleground between parents and offspring, empirical studies in animals focus disproportionately on time or energy investments by parents. Water is another vital and limiting resource that, contrary to energy, is rarely stored. Compromised water balance (i.e. dehydration) usually results in lower locomotor performances (i.e. endurance, jumping speed) (Prates, Angilleta, Wilson, Niehaus, & Navas, 2013) and reduced activity (Davis & DeNardo, 2009; Lorenzon, Clobert, Oppliger, & John-Alder, 1999), although positive effects may also occur such as increased immune defences (Brusch & DeNardo, 2017; Moeller, Butler, & DeNardo, 2013) or a boost of antioxidant defences (Stier et al., 2017). During reproduction, water is essential for both the mother and embryos (e.g. in humans, Hirschhorn, Chowdhury, & Lindenbaum, 1969; Cheung & Lafayette, 2013) but studies that examine non-energetic allocation constraints for water and the potential for water limitations to induce inter-generational trade-offs between mother and offspring are exceedingly rare (Dupoué, Brischoux, et al., 2015).

Several lines of evidence suggest intergenerational trade-offs and conflicts in parental water investment in regards to the effects of water availability on reproductive effort and reproductive success in a wide diversity of animal species (Coe & Rotenberry, 2003; Lecomte, Gauthier, & Giroux, 2009; Ross & Desai, 2005). Yet, it is often complicated to distinguish the effects induced by energetic trade-offs from those due to water limitation when these resources are concomitantly provided to the offspring (e.g. during lactation in mammals, during food provisioning in altricial birds, or during egg production). Regular reliance on stored energy (i.e. capital breeding) make thus several viviparous species of squamate reptiles (i.e. lizards and snakes) excellent study systems to advance our understanding of the effects of water limitation on reproduction. Most viviparous squamates are lecithotrophic so that energy supply is prominently provided during vitellogenesis before ovulation (Blackburn, 2000). During pregnancy, maternal physiological effort is mainly oriented into an investment of water, oxygen and some inorganic elements, and offspring needs for water to convert yolk into somatic tissue results in exponential water requirements over pregnancy (Bonnet, Naulleau, & Shine, 2017; Lourdais, Lorient, Dupoué, Wright, & DeNardo, 2015; Vleck, 1991). This temporal dissociation between energy and water allocation enables one to focus on factors shaping maternal water investments without confounding energy effects.

Whenever water becomes limited in viviparous squamate reptiles, maternal water needs may trade with embryo requirements (Dupoué, Brischoux, et al., 2015) and induce maternal physiological stress (as suggested by increased plasma corticosterone levels; Romero, Dickens, & Cyr, 2009) and sibling competition (Bonnet et al., 2017). We found in a previous study that water-deprived pregnant asp viper (*Vipera aspis*) experienced a compromised hydration state in favour of unaltered offspring development (Dupoué, Brischoux, et al., 2015) and differential offspring growth early in life in the laboratory, suggesting a potential adaptive response to water deprivation (Dupoué et al., 2016). However, asp viper are nearly semelparous (Bonnet, Naulleau, Shine, & Lourdais, 1999), so compromised female state likely has minimal impact on fitness. It is therefore critical to measure the long-term effects of water limitation on females as well as offspring fitness indicators such as the timing of the next reproduction, annual juvenile growth rate, and juvenile survival rate. Such estimates are crucial for investigating the occurrence and outcome of mother-offspring conflicts in more typical reproductive strategies.

Accordingly, in this study, we aimed to determine the extent to which water restriction during gestation impacts female physiological indicators, reproductive output, future reproduction and survival in the iteroparous Common Lizard (*Zootoca vivipara*). To do so, we exposed pregnant females and males to a 2-week period of water restriction during the latter half of gestation. Intensity and duration of water restriction were designed to fit with ecologically relevant levels of constraint (Lorenzon et al., 1999). The direct physiological changes induced by water restriction were quantified by measuring two traits related to water balance (body mass and plasma osmolality) and one related to physiological stress (plasma corticosterone level). Changes in body mass are related to hydration state in reptiles (DeNardo, Zubal, & Hoffman, 2004; Lillywhite, Babonis, Sheehy, & Tu, 2008), and plasma osmolality is the most relevant indicator of dehydration in vertebrates lacking salt glands. In addition, plasma corticosterone level generally increases in response to acute and chronic environmental stressors in order to elicit adaptive physiological and behavioural responses, such as shifts in metabolic pathways or immune defences (Romero et al., 2009). In reptiles, corticosterone may also be involved in the regulation of the hydro-mineral balance (i.e. mineralocorticoid function; McCormick & Bradshaw, 2006). Physiological responses in females were compared with those of males of the same age and under the same environmental conditions, and compared across a gradient of variation in reproductive effort among individuals. Greater dehydration and higher corticosterone levels were expected in pregnant females compared to males and in females with increasing fecundity. In addition, we analysed the effects of water restriction on female reproductive traits, including parturition date and reproductive output, and on offspring traits (offspring size and condition at birth and early juvenile growth rate). We predicted delayed reproduction and decreased offspring quality due to water restriction as a consequence of less efficient energy conversion from yolk to embryonic tissues. Finally, we recorded 1-year growth and survival of mothers and their offspring, as well as reproductive output by females during the following year in an attempt to reveal long-lasting fitness effects.

2 | MATERIALS AND METHODS

2.1 | Study species and housing conditions

The European Common Lizard, *Zootoca vivipara*, is a small (adult snout-vent length [SVL] c. 50–75 mm), widespread lacertid from cold, humid peat bog and heathland habitats across northern Eurasia. Natural populations are structured into three age classes: juveniles (newborn individuals), yearlings (1–2 years old) and adults (2 years or more). At our study site, males start to emerge from hibernation around the beginning of March, followed shortly thereafter by juveniles and then females a few weeks later (Rutschmann et al., 2016). From June to July, females lay an average clutch of five non-calcified eggs (range 1–12). In our study area, the climate and food conditions experienced from birth were sufficiently favourable to enable most of the yearling females to initiate reproductive activity. Offspring hatch a few minutes after parturition and are immediately autonomous. Hibernation begins in October.

Between 18 and 27 May 2015, we caught 185 adult females, 90 adult males, 47 yearling females and 71 yearling males from 24 outdoor enclosures (10 × 10 m) at the Centre de Recherche en Ecologie Expérimentale et Prédictive (Saint-Pierre-lès-Nemours, France, 48°17'11.42N, 2°40'46.00E). These lizards were previously used in a field experiment in that differed in water access, availability of basking and refuge sites, and vegetation cover, since these are critical determinants of population density in the wild. In “control” enclosures, we used our standard microhabitat set-up and maintenance procedures with two permanent watering holes, four piles of stones and logs used as shelters and basking sites by lizards, and tall, dense vegetation. In the “low quality” enclosures, watering holes, logs and stones were removed, and grass was regularly cut to maintain a maximum vegetation height of 20 cm. To disentangle effects of vegetation cover from effects of microhabitat diversity and water availability, we further produced “intermediate quality” enclosures where grass was not manipulated but all watering holes, logs, and rocks were removed (see details in Josserand et al., 2017). For this study, all enclosures were restored to the control treatment immediately after the initial capture of all lizards.

As vitellogenesis occurred prior to the initial capture and most maternal investment of energy occurs during vitellogenesis (Bleu et al., 2013), initial enclosure habitat quality had a positive effect on fecundity, likely caused by varying food resources among enclosure types and the effect of food availability on maternal energetic investment (Josserand et al., 2017). However, since females were randomly assigned to the treatment groups of this study, we did not find any additive or interactive effects of habitat quality on the physiological responses or life-history traits in our study (results not shown). Therefore, effects of habitat quality were discarded from our analyses and will not be discussed.

Each lizard was identified with a unique toe clip code, measured (SVL, ±0.5 mm) and weighed (body mass [BM], ±1 mg). Lizards were individually housed in terraria (adults: 25 × 15 × 16 cm, yearlings: 18 × 11 × 12 cm) containing a shelter, peat soil as substrate and

opportunities for optimal thermoregulation created by suspending a 25 W light bulb over one end of the terraria. Light was sequentially on for 9 h/day between 08.00 and 19.00 to provide a daytime thermal gradient of 23°C at the cool end and 35–38°C below the bulb. From 19.00 to 8.00, the cages were at room temperature, 17°C. All lizards had access to a standardized mass of crickets (*Acheta domesticus*) every 3 days (adults: 400 ± 20 mg, yearlings: 300 ± 20 mg). Lizards had ad libitum access to a water bowl, and terraria were sprayed with water three times per day (09.00, 13.00, 17.00). Pregnant females were maintained in captivity until parturition. Immediately after parturition, the juveniles were separated from their mother, marked by toe clipping and measured. Adult males and yearlings were released a few days after the experiments.

2.2 | Experimental design

At the time when pregnant females were at mid-gestation (1–15 June), we randomly assigned lizards within each age class and sex to two experimental treatments following a previously established procedure to contrast both the air humidity and the access to free standing water (Lorenzon et al., 1999). In the water-restricted treatment, we removed the water bowl and reduced the misting frequency to once per day occurring at 09.00 (adults, females: $n = 95$, males: $n = 47$; yearlings, females: $n = 25$, males: $n = 35$). In the control treatment, lizards had permanent access to the water bowl and were misted three times per day (adults, females: $n = 90$, males: $n = 43$; yearlings, females: $n = 22$, males: $n = 36$). The control treatment mimics conditions in which lizards find permanent access to water (e.g. peat bog, marsh), whereas the water-restricted treatment reflects drier conditions (e.g. heath, dry meadows) in which, during summer, morning dew is the only source of free water. Water restriction lasted for 14 days, which, on average, occurred from 33 to 19 days before parturition, a time when energy allocation into offspring is negligible. After the period of water restriction, all individuals returned to the control water conditions, having permanent access to water in a water bowl and being misted three times per day.

2.3 | Measurements of water balance and stress

Individuals were weighed (±1 mg) every 3 days throughout the water restriction period; however, for simplicity, we only examined BM change (Δ BM) from the initiation to the end of the water restriction period. We also measured changes in plasma osmolality (Δ OSMO) and plasma concentrations of corticosterone, the primary glucocorticoid in reptiles (Δ CORT). For these assessments, lizards were bled within 3 min of handling at the onset and the end of the water restriction period using a standard protocol (Meylan, Dufty, & Clobert, 2003). Blood samples (adults: c. 60 µl; yearlings: c. 40 µl whole blood) were collected from the post-orbital sinus and centrifuged for 5 min at 13,500 × g. Plasma was then separated from the blood cells and kept frozen at –30°C in airtight tubes until analyses were performed. In adults, plasma was divided into two 10 µl aliquots, allowing for

the measurement of both osmolality and corticosterone (see below), whereas the volume of plasma collected from yearlings restricted analysis to only corticosterone.

Plasma osmolality (± 1 mOsm/kg) was determined using a vapour pressure osmometer (model 5,500, Wescor, Logan, UT, USA) according to the protocol described in Wright, Jackson, and DeNardo (2013). Before analyses, plasma was diluted (1:1) in reptile Ringer's solution (300 mOsm/kg) prepared following methods from Secor, Stein, and Diamond (1994) so that plasma osmolality could be determined from 10 μ l duplicates (intra-individual coefficient of variation: 3.9%). The osmolality of each blood sample was calculated as the average osmometer reading times 2 – 300. Corticosterone was measured with a competitive enzyme immunoassay (IDS Corticosterone EIA kit, ref AC-14F1, Immunodiagnostic Systems Ltd, France) after a 1:10 dilution of all samples. This method quantifies total plasma corticosterone using a polyclonal corticosterone antibody and is based on a highly repeatable colorimetric assay of absorbance at 450 nm (12 plates with four replicates per plate: intra-plate coefficient of variation: 17% and inter-plate coefficient of variation: 31%; intra-plate repeatability: $r = .98$, $F_{1,34} = 168.7$, $p < .001$; inter-plate repeatability: $r = .81$, $F_{11,34} = 9.6$, $p < .001$, Lessells & Boag, 1987).

2.4 | Measurements of reproductive effort and offspring quality

We examined the effects of water restriction on different reproductive traits. We compared parturition date and counted and weighed (± 1 mg) live juveniles, dead juveniles and undeveloped eggs of each litter to calculate total litter size and mass (i.e. number and mass of all components) as well as fit litter size and mass (i.e. number and mass of live juveniles only). For live juveniles from both water-restricted and control mothers, we measured BM (± 1 mg), SVL (± 1 mm) and body condition (BC, estimated as the residuals derived from the linear relationship between BM and SVL, $t_{1,1613} = 40.2$, $p < .001$, $r^2 = .50$).

One to 2 days after birth, siblings were released as a group into one of the 24 outdoor enclosures, whereas the mother was released into a different enclosure. Enclosure assignments were made randomly, but adjusted as needed so that enclosures contained the same number of lizards from each sex, age class and water treatment. This experimental set-up enabled us to examine the effects of water restriction while limiting confounding effects from other variables. From 7 to 11 September 2015, we recaptured a subset of juveniles ($n = 390$) from all enclosures and re-measured SVL. We determined early juvenile growth rate, as the change in SVL from birth divided by the duration between measurements (in days).

2.5 | Measurements of annual growth, survival and future reproduction

We estimated the long-term effects of water restriction on annual growth rate, annual survival, and next reproductive effort in females. We recaptured lizards from the enclosures a year after the water

restriction experiment (16 May–3 June 2016). On the day of capture, lizards were identified by their unique toe-clip code, measured, weighed and then housed in terraria. We were able to estimate lizard survival with a high degree of reliability because, in these enclosures, recapture probability is close to 1 (Le Galliard, Fitze, Ferrière, & Clobert, 2005). Annual growth rate was calculated as the change in SVL divided by the days between measurements, since growth rate in this species is almost linear over months (Fitze & Le Galliard, 2008). However, we subtracted the number of days when lizard are physiologically inactive; that is, we assumed a period of 120 days between the onset of hibernation around the end of October and emergence in early March. Reproductively active females were further monitored until parturition to assess their next reproductive investment. To do so, we measured the same indicators of reproductive investment as described above.

2.6 | Statistical analyses

All analyses were performed using R software (version 3.2.0, R Development Core Team 2016, <https://www.r-project.org/>). Changes in physiological traits directly related to water balance (Δ BM and Δ OSMO) and stress (Δ CORT) were analysed using linear models. For Δ BM and Δ CORT analyses, models included fixed effects of treatment, age class, sex and their first-order and second-order interactions. Because osmolality was measured only in adults, models of Δ OSMO included fixed effects of treatment, sex and their first-order interactions. We added the effects of initial physiological value (at the onset of water restriction) as linear covariates. In adults, we also examined the relationships between Δ BM and both Δ CORT and Δ OSMO using linear models.

Reproductive traits (parturition date and reproductive output) were analysed with linear mixed models with the enclosure from where the female was captured as a random effect (package NLME, Pinheiro, Bates, DebRoy, Sarkar, & R Core Team, 2016). Female age class, treatment and the first-order interaction were included as fixed effects. Offspring morphology and early juvenile growth rate were analysed with similar models except that random factors included mother identity (sibling non-independence) nested in the enclosure into which the juveniles were released. In all models, random effects strongly influenced the response variables (likelihood ratio tests, all $p < .001$).

Long-term effects of water restriction on annual growth and female reproductive effort were also analysed with linear mixed models, with enclosure into which the lizard was released as a random factor (likelihood ratio tests, all $p < .001$). Annual survival was analysed with mixed-effect logistic regressions including a logit link and binomial error term (package LME4, Bates, Maechler, Bolker, & Walker, 2015) and z tests (package LMERTEST, Kuznetsova, Bruun Brockhoff, & Haubo Bojesen Christensen, 2016). We conducted separate analyses for juveniles (offspring born from manipulated mothers), yearlings and adults from the laboratory manipulation. In adults and yearling models, the enclosure into which they were released was used as a random effect and treatment, age class and

sex were treated as fixed effects together with their first-order and second-order interactions. Initial SVL was used as a covariate to control for a potential body size effect on survival and growth. In juvenile analyses, sex, mother age class and mother treatment and their interaction were included as fixed factors, whereas mother identity and the enclosure into which the juvenile was released were treated as additive random factors (likelihood ratio tests, all $p < .001$). In all cases, a minimum adequate model was obtained by a backward procedure where we removed non-significant terms one by one starting with second-order interactions. Results are presented as the mean $\pm$ standard error.

3 | RESULTS

3.1 | Physiological responses to water restriction

At the onset of the experiment, BM, SVL and BC of water-restricted individuals did not differ significantly from their respective sex and age class controls (all $p > .295$). At the end of the water restriction, Δ BM was not related to initial BM ($t_{1,370} = -1.43$, $p = .153$) and was significantly influenced by hydric treatment depending on age class and sex (three-way interaction: $t_{1,370} = -2.62$, $p = .009$). Water restriction strongly reduced the increase in BM of adult pregnant females (contrast term: $\beta = -0.766 \pm 0.054$ g, $t_{1,370} = -14.09$, $p < .001$) compared to other groups including adult males ($\beta = -0.162 \pm 0.075$ g, $t_{1,370} = -2.15$, $p = .032$), yearling females ($\beta = -0.241 \pm 0.107$ g, $t_{1,370} = -2.26$, $p = .024$) and yearling males ($\beta = -0.069 \pm 0.085$ g, $t_{1,370} = -0.81$, $p = .416$, see Figure 1a). In adults, Δ OSMO at the end of the experiment was negatively related to initial osmolality ($t_{1,199} = -15.41$, $p < .001$) and was greater in water-restricted individuals compared to controls ($\beta = 19.29 \pm 5.57$ mOsm/kg, $t_{1,199} = 3.47$, $p < .001$, Figure 1b), regardless of the sex or the interaction between sex and treatment (all $p > .449$, Figure 1b).

Δ CORT was negatively related to initial CORT ($t_{1,354} = -15.12$, $p < .001$), and was higher in water-restricted individuals ($\beta = 17.93 \pm 6.32$ ng/ml, $t_{1,354} = 2.84$, $p = .004$, Figure 1c) regardless of other factors (all $p > .220$). Mean Δ CORT was significantly higher in pregnant females compared to males ($\beta = 27.05 \pm 0.036$ ng/ml, $t_{1,354} = 3.77$, $p < .001$, Figure 1c). In females, Δ BM was negatively related to Δ OSMO in both water-restricted and control individuals ($t_{1,134} = -2.98$, $p = .003$), whereas Δ CORT did not correlate with Δ OSMO in either water-restricted or control lizards ($t_{1,130} = 0.50$, $p = .621$). In males, Δ BM negatively correlated with Δ OSMO in water-restricted individuals ($t_{1,62} = -2.90$, $p = .005$) but not in controls ($t_{1,62} = -0.09$, $p = .929$), and there was no correlation between CORT and OSMO ($t_{1,63} = -1.10$, $p = .277$).

Water-restricted females with higher fecundity experienced greater dehydration, with Δ BM negatively ($t_{1,213} = -3.13$, $p = .002$, Figure 2a) and Δ OSMO positively ($t_{1,133} = 2.30$, $p = .023$, Figure 2b) correlated with fecundity, whereas no correlations with Δ BM and Δ OSMO were found in control females (all $p > .718$). There were no correlations between Δ CORT and fecundity in either water-restricted or control females (all $p > .839$).

3.2 | Effects of water restriction on reproductive output

Overall, adult females gave birth sooner ($\beta = 10.08 \pm 0.94$ days, $t_{1,192} = 10.72$, $p < .001$) and were more fecund than yearlings (all $p < .001$). However, none of the reproductive traits were impacted by water restriction or by the interaction between water restriction and age class (all $p > .225$). That is, water restriction had no significant effect on parturition date (control: 6 July $\pm$ 0.6 days, water-restricted: 6 July $\pm$ 0.7), total litter size (control: 7.57 $\pm$ 0.23, water-restricted: 7.52 $\pm$ 0.21), total litter mass (control: 1.405 $\pm$ 0.048 g, water-restricted: 1.393 $\pm$ 0.052 g), fit litter size (control:

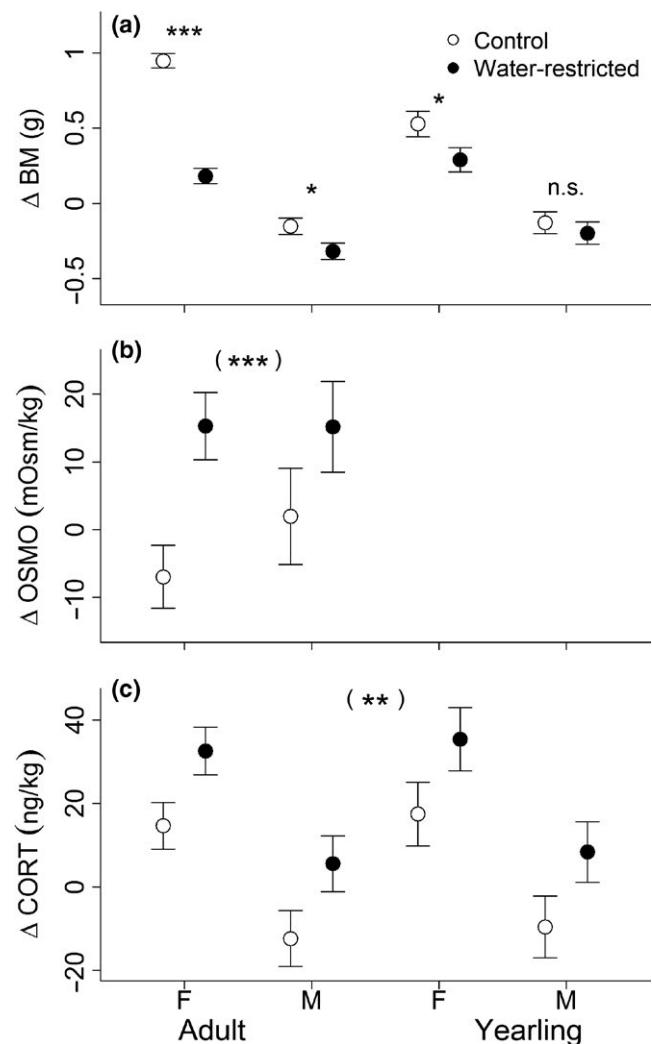


FIGURE 1 Effect of water restriction on changes in physiological indicators in pregnant female and male common lizards (*Zootoca vivipara*) from two age classes. Points represent the least square means ($\pm$ SE) adjusted to initial values of (a) the change in body mass (Δ BM), (b) the change in plasma osmolality (Δ OSMO), and (c) the changes in plasma corticosterone (Δ CORT), in control (open circles) and water-restricted individuals (filled circles). Significant effects of water restriction within age class and sex are symbolized: * $p < .05$, ** $p < .01$, *** $p < .001$ and n.s. non-significant, whereas the overall effects of water restriction are symbolized: (*) $p < .01$, (***) $p < .001$

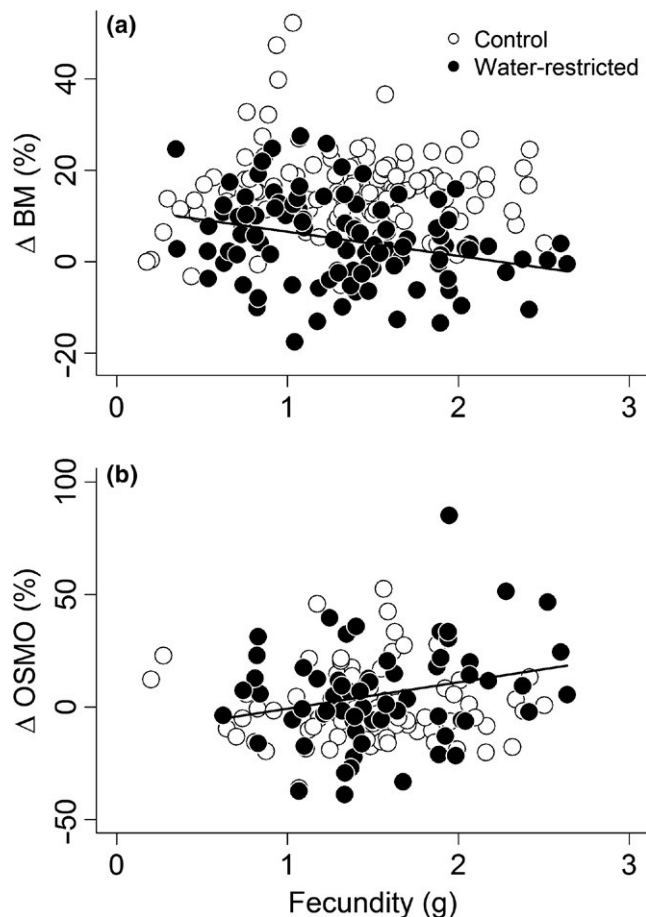


FIGURE 2 Relationships between the relative change in physiological indicators of hydration state and the total litter mass of developing embryos. In water-restricted pregnant females, (a) The change in body mass (ΔBM) was negatively related to fecundity ($p = .002$, $r^2 = .086$) and (b) the changes in plasma osmolality ($\Delta OSMO$) was positively related to fecundity ($p = .023$, $r^2 = .063$), whereas no relationship was found in control females (all $p > .718$)

7.30 ± 0.25 , water-restricted: 7.18 ± 0.22) or fit litter mass (control: 1.362 ± 0.050 g, water-restricted: 1.341 ± 0.052 g). Similarly, water restriction during pregnancy had no significant effect on juvenile SVL (control mothers: 20.633 ± 0.045 mm, water-restricted mothers: 20.627 ± 0.043 mm, $t_{1,207} = 0.45$, $p = .655$), BM (control: 186.1 ± 1.0 mg, water-restricted: 186.5 ± 1.1 mg, $t_{1,210} = 0.20$, $p = .843$) and BC at birth (controls: $-2.00 \times 10^{-4} \pm 2.07 \times 10^{-4}$, water-restricted: $7.02 \times 10^{-4} \pm 7.65 \times 10^{-4}$, $t_{1,207} = 0.07$, $p = .945$).

3.3 | Effects of water restriction on growth and survival of yearlings and adults

Water restriction had no effect on female or male, adult or yearling annual body growth rate (all $p > .248$), and overall, males tended to grow faster than females ($\beta = 3.20 \pm 1.66 \mu\text{m/day}$, $t_{1,65} = 1.93$, $p = .058$, Figure 3a), and adult growth rate was lower than yearling growth rate ($\beta = 16.21 \pm 1.55 \mu\text{m/day}$, $t_{1,65} = 10.45$, $p < .001$). Similarly, water restriction had no effect on survival rate (all $p > .646$, Figure 3b),

but overall, females had better survival than males ($\beta = 0.57 \pm 0.27$, $z = 2.15$, $p = .032$) and adults had lower survival than yearlings ($\beta = -0.58 \pm 0.26$, $z = -2.25$, $p = .025$). In surviving females, those that were exposed to water restriction during their previous pregnancy had higher subsequent year total litter mass ($\beta = 0.227 \pm 0.107$ g, $t_{1,30} = 2.13$, $p = .042$) and fit litter mass ($\beta = 0.235 \pm 0.115$ g, $t_{1,30} = 2.04$, $p = .050$, Figure 3c) than did females that were controls during their previous pregnancy.

3.4 | Effects of maternal water restriction on offspring growth and survival

Early growth rate was not related to initial SVL ($t_{1,211} = -1.15$, $p = .253$) but was significantly impacted by the interaction between juvenile sex and mother treatment ($t_{1,211} = 2.77$, $p = .006$, Figure 3d). Early growth rate was higher in female offspring from control mothers compared to females from water-restricted mothers ($\beta = 17.17 \pm 6.27 \mu\text{m/day}$, $t_{1,151} = 2.74$, $p = .007$, Figure 3d), whereas mother treatment had no effect on male early growth rate ($t_{1,151} = 0.34$, $p = .731$, Figure 3d). Annual growth rate of juveniles was neither related to initial SVL ($t_{1,104} = -1.11$, $p = .270$) nor impacted by mother treatment ($t_{1,99} = 0.25$, $p = .801$, Figure 3e), but was greater in females ($\beta = 4.18 \pm 2.03 \mu\text{m/day}$, $t_{1,104} = 2.06$, $p = .042$, Figure 3e). Annual survival was positively related to initial SVL ($z = 4.94$, $p < .001$) and was lower in juveniles from water-restricted mothers compared to control mothers ($\beta = -0.35 \pm 0.16$, $z = -2.08$, $p = .038$, Figure 3f). Annual survival of juveniles was positively correlated with their early growth rate ($\beta = 7.92 \pm 2.40$, $z = 3.30$, $p < .001$, Figure 3g).

4 | DISCUSSION

As a result of water restriction, individuals showed greater dehydration and increased physiological stress on average, which is consistent with general expectations and results from our previous works in a viviparous, semelparous snake (Dupoué, Brischoux, et al., 2015; Dupoué et al., 2016). However, the previous study showed that, at least in the short-term, intergenerational water-based trade-offs may be solved in favour of the embryos. Here, in a long-term monitoring study of an iteroparous species, we show contrasting effects in that water deprivation during pregnancy had greater long-term impacts on the offspring than on the female. That is, while water restriction did not negatively impact current reproductive output, offspring from water-restricted mothers had lower annual survival and water-restricted mothers had enhanced future reproductive investment. These long-term fitness responses to water restriction in offspring and mothers suggest that water can trigger intergenerational conflicts as have been demonstrated for energy (Godfray et al., 1991; Hinde et al., 2010; Kölliker et al., 2015; Parker & Mock, 1987).

Pregnant females exposed to water restriction had greater BM loss relative to control females compared to water-restricted adult males or yearlings and their controls, possibly suggesting a higher sensitivity to water restriction during gestation. Yet, the increase in plasma

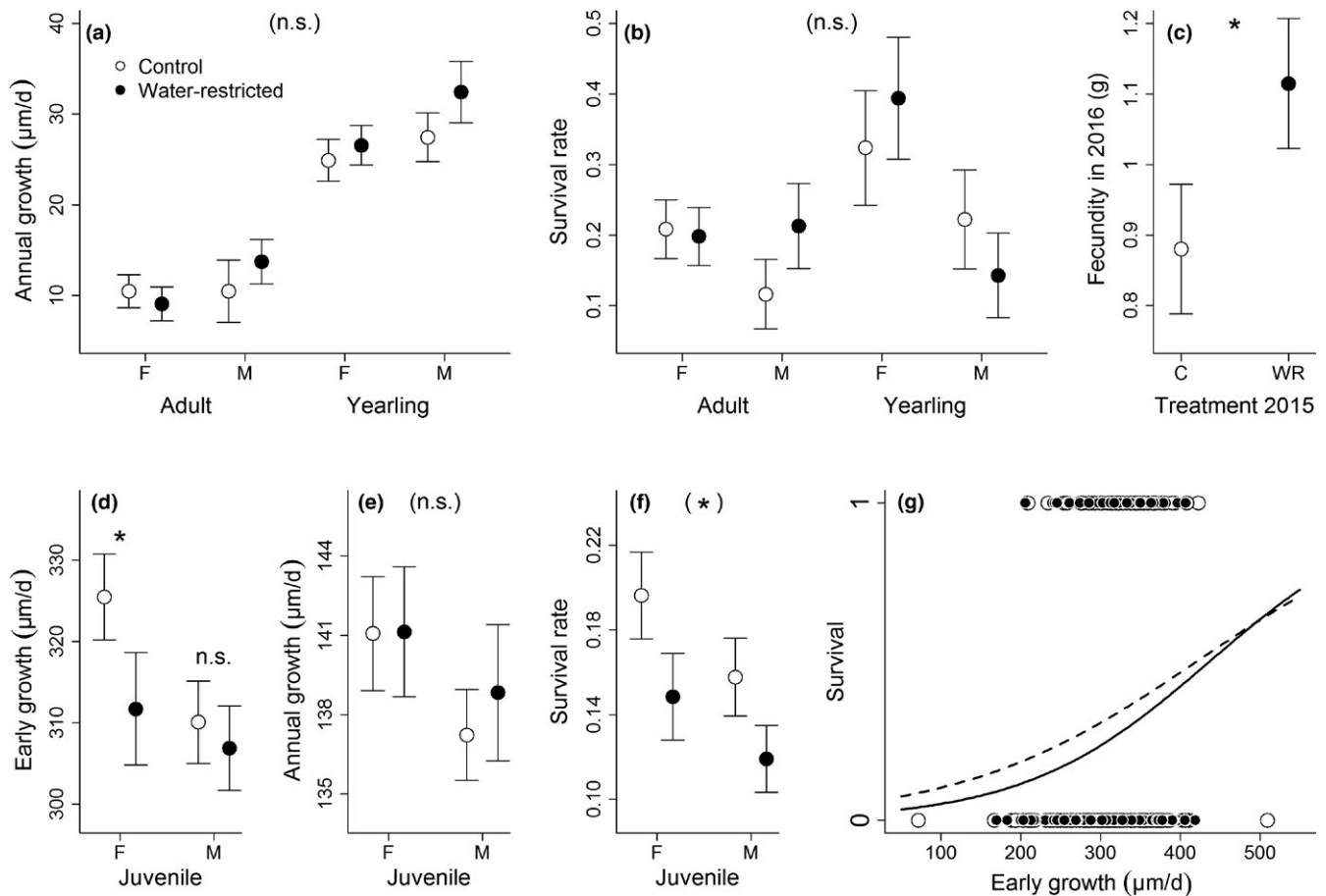


FIGURE 3 Long-term effects of water restriction on fitness estimates. Top panels (a–c) represent fitness estimates in adults and yearlings, whereas bottom panels (d–g) represent effects in juveniles. Points represent the $M (\pm SE)$ of control (open circles) and water-restricted individuals (filled circles) in adult (a) annual growth rate, (b) survival rate, (c) next reproductive effort, and juvenile (d) early growth rate, (e) annual growth rate, and (f) survival rate. (g) The positive relationship between early growth and survival to 1 year (control: dashed line, water restricted: solid line). Significant differences between control and water restricted individuals within age class and sex are symbolized: * $p < .05$, and n.s., non-significant, and the overall effect of water restriction is symbolized: (*) $p < .05$, and (n.s.) non-significant

osmolality was similar between adult females and males, despite expecting pregnant females to face greater changes in osmolality, since pregnancy increases both total water loss (Dupoué, Stahlschmidt, Michaud, & Lourda, 2015; Webber, Gibbs, & Rodríguez-Robles, 2015) and water demand (Cheung & Lafayette, 2013; Lourda et al., 2015). However, it is important to realize that, in our study, adult females differed from adult males in both reproductive state and sex (i.e. there were no non-reproductive females), making it impossible to clearly quantify the water cost of pregnancy *per se*. Besides, water-restricted individuals were provided with some water in the morning so that pregnant females could adjust drinking behaviour and activity patterns to compensate for relatively higher water loss or investment. Regardless, greater reproductive output led to greater dehydration state in water-restricted females but not in control females, suggesting a consequential water demand associated with a higher reproductive investment (Dupoué, Brischoux, et al., 2015). This last result demonstrates that, whenever availability of water is limited, there is a trade-off for the allocation of water between mothers and their offspring similar to the oft-mentioned allocation trade-off for energy.

Pregnancy resulted in higher corticosterone levels and higher corticosterone changes which are consistent with previous work (Dauphin-Villemant & Xavier, 1986; Lorigou, Angelier, & Lourda, 2016; Taylor, DeNardo, & Jennings, 2004). However, despite relatively high intra- and inter-plate coefficients of variation, we found that corticosterone increased similarly in all water-restricted individuals regardless of age class and sex, likely linking this increase to stress imposed by dehydration. We did not find any relationship between changes in corticosterone and changes in mass or osmolality, suggesting a complex function of corticosterone in the regulation of water balance as has been previously suggested (Dupoué, Angelier, Lourda, Bonnet, & Brischoux, 2014; Dupoué et al., 2016; McCormick & Bradshaw, 2006).

While pregnant females experienced multiple physiological effects of water restriction, their immediate reproductive output and offspring quality were buffered from effects of the water restriction. It is noteworthy that females were not completely water deprived and, furthermore, they were all provided *ad libitum* water access for almost 3 weeks before parturition. It is therefore possible that females corrected any negative effects of the water restriction during

this pre-parturition ad libitum access to water. It is also possible that, during the period of water restriction, embryo water balance might be protected by extra-embryonic fluids, although the value of this water resource is rarely evaluated (Bonnet et al., 2017). While such water storage would be valuable for embryo water balance, it might entail costs for both mothers (e.g. greater physical burden) and embryos (e.g. lower nutrition or oxygenation) and, therefore, should be further investigated. We did not find any effects of water restriction on adult annual growth or survival rates. Rather, juveniles from mothers that were water-restricted had lower annual survival than those from control mothers despite no difference in their body size or condition at birth. Several, non-mutually exclusive explanations may have contributed to these long-term, rather than short-term, impacts on juveniles (Mugabo, Marquis, Perret, & Le Galliard, 2011). First, other physiological and behavioural attributes of offspring at birth may contribute to their lower expectations of future survival. For example water restriction led to elevated plasma corticosterone levels in pregnant females, and elevated corticosterone during embryonic development is often associated with lower offspring quality. However, the net longer term effects of corticosterone secretion during gestation on juvenile growth and survival rates are complex with both positive or negative effects reported in the different studies (Bonier, Martin, Moore, & Wingfield, 2009; Dantzer et al., 2013; Gagliano & McCormick, 2009; Meylan & Clobert, 2005). More detailed mechanistic examinations of how water restriction may impact maternal effort and thus influence offspring development and long-term fitness are greatly needed, especially investigations of unexplored maternal physiological (e.g. arginine vasotocin, oxidative stress) and behavioural (e.g. selected temperature, drinking behaviour) traits that could negatively impact juvenile performance (e.g. immunity, physiological stress) and thereby survival without influencing the limited offspring traits we examined (Mousseau, Uller, Wapstra, & Badyaev, 2009).

Equally interesting and surprising, water-restricted females invested in greater reproductive effort the next year. Energy-based trade-offs between current and future reproductive investment have been previously demonstrated (Harshman & Zera, 2007; McNamara & Houston, 1996; Senner et al., 2015). However, in our water-based assessment, current reproductive output (e.g. parturition date, offspring number/quality) was not altered by water restriction. Whether the reduced short-term growth and annual survival of offspring reflect a water-based maternal trade-off between current and future reproductive efforts independent of energy is of considerable interest. Alternatively, increased reproductive effort the year after water restriction during pregnancy may result from poor conditions experienced by females independent of water-based trade-offs, as has been frequently illustrated, for example in studies of terminal investment (Creighton, Heflin, & Belk, 2009; Velando, Drummond, & Torres, 2006). Interestingly, higher plasma corticosterone concentrations during reproduction, as we saw in water-restricted female Common Lizards, have been shown to increase future reproductive investment (Love & Williams, 2008). The effects of water availability on the trade-off between current and future reproduction should be specifically examined as such trade-offs are typically explained from energy allocation

rules and are evaluated based on energetic measures of reproductive effort and allocation into offspring quality (e.g. mass of the litter, mass of offspring, and energy expenditure).

Altogether, our experiment suggests that a conflict of water allocation between mother and offspring may exist whenever the access to water resource is compromised. Antagonistic co-evolution between mother and offspring occurs between genes shaping maternal investment whenever the optimum for resources differ (Godfray et al., 1991; Hinde et al., 2010; Kölliker et al., 2015; Parker & Mock, 1987). Although our study does not allow to test for genetic trade-offs between offspring and mothers, it is suggestive that mothers could regulate water transfer to favour individual fitness at the expense of their offspring performance. Future studies should therefore examine physiological and genetic mechanisms involved in water transport during pregnancy. For instance on the female side, pregnancy can upregulate the expression of some genes related to water allocation (e.g. aquaporin genes in the uterus of pregnant skink *Chalcides ocellatus* uterus; Brandley, Young, Warren, Thompson, & Wagner, 2012). On the embryo side, embryos may actively extract and store water from external or maternal environments by upregulating genes supporting water transfer through the chorioallantoic membranes (Griffith, Brandley, Whittington, Belov, & Thompson, 2017; Shine & Thompson, 2006).

Intergenerational conflicts for water likely provide a selective pressure in the evolution of optimal maternal investment as is seen for energy (Godfray et al., 1991; Hinde et al., 2010; Kölliker et al., 2015; Parker & Mock, 1987). For instance, previous work has shown that clutch size negatively correlates with environmental aridity in birds (Oppel et al., 2013; Tieleman, Williams, & Visser, 2004). Because aridity is also associated with lower food availability, it is difficult to decouple the effects of water from those of energy availability on maternal investment. Still, water supplementation experiments in desert (black-throated sparrow, *Amphispiza bilineata*) and arctic (greater snow geese, *Chen caerulescens atlantica*) birds demonstrated that water is a determinant resource in shaping current reproductive investment (Coe & Rotenberry, 2003; Lecomte et al., 2009). Intergenerational conflicts for water may also provide new insights in understanding the transition to viviparity (Bonnet et al., 2017; Crespi & Semeniuk, 2004; Pollux et al., 2014; Schrader & Travis, 2008). Here and according to our results, we hypothesize that water availability in the environment should limit this transition because viviparity implies a stronger reproductive investment in water. In support of this, the proportion of viviparous species decreased in arid areas in most continents (Feldman et al., 2015), whereas it is more likely to appear in tropical humid regions (Lambert & Wiens, 2013). More investigations are required to clarify the water constraints of prolonged egg retention and the influence of water availability on transition to viviparity.

Intergenerational trade-offs and conflicts for water are likely broadly relevant. Even in plants, for instance, seed and fruit formation often requires a considerable water investment and, in xeric environments, may result in parent-offspring conflicts (Uma Shaanker & Ganeshaiah, 1997). Water resources may also shape postnatal parental care, yet such a possibility remains largely overlooked (Tu, Lillywhite, Menon, & Menon, 2002). Species from arid environments with specific

strategies for provisioning offspring with water (e.g. water transport in sandgrouse plumage; Mougeot, Benítez-López, Casas, García, & Viñuela, 2014) could be particularly valuable in revealing water-based parental costs. Eventually, the clarification of parent–offspring conflicts for water will be critical in understanding general pattern such as habitat use, demographic trends and predicting general responses to climate changes.

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AUTHORS' CONTRIBUTIONS

This study was designed by A.D., J.-F.L.G., and S.M. Experiments were led by A.D. and data were collected by all authors. A.D. led the data analyses together with J.-F.L.G. and S.M. All authors contributed in result interpretation and manuscript writing.

CONFLICT OF INTEREST

The authors declare no competing or financial interests.

DATA ACCESSIBILITY

Data available from the Dryad Digital Repository <https://doi.org/10.5061/dryad.s074j> (Dupoué et al., 2017).

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